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Stored Procedure

A stored procedure is a collection of SQL statements saved inside a database that we can execute whenever we need them.

Think of it like a saved recipe. Instead of writing the same SQL query again & again, we save it once and call it when ever we want.

for ex, if we frequently need to display all employees, we can create a stored procedure for that task.

Why do we use stored procedures?

- Reusability: write SQL code once and reuse it
- Performance: Can reduce repeated processing and improve efficiency in suitable cases
- Security: Access can be granted to execute a procedure without granting direct aces to all underlaying tables.
- Reduce network traffic: Send a procedure call instad of sending a long SQL query each time
- Easy maintenance: update the logic in one place

Page No.: ---

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Syntax : DELIMITER //

```
CREATE PROCEDURE procedure_name()
BEGIN
    SELECT * FROM emp;
END //
DELIMITER ;
```

DELIMITER // : temporarily changes the statement ending symbol to //

CREATE PROCEDURE : creates a stored procedure

procedure_name() : Name of the procedure; here no parameters are used

BEGIN : starts the procedure body

SELECT * FROM emp; : retrieves all column of rows from the emp table

END // : Ends the procedure definition

DELIMITER ; : Restores the normal semicolon delimiter

DELIMITER : Normally, MySQL uses ; to identify the end of the statement. But a procedure can contain multiple SQL statements, each ending with a semicolon,

So we temporarily change the delimiter to //, allowing MySQL to recognize the end of the

Page No.: _____

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entire procedure definition

step 1 - create

Ex: DELIMITER //

```
CREATE PROCEDURE getEmployees()
BEGIN
    SELECT * FROM emp;
END //
DELIMITER ;
```

step 2 : Execute :

```
CALL getEmployees();
```

Types of parameters

A parameter allows us to pass information into a procedure or get information back from it

1. IN : Input parameter , we give a value to procedure and it uses that value

Ex: Find the employee whose employee no is 7521

```
DELIMITER //
CREATE PROCEDURE getEmployeeById (IN eid
INT)
```


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```
BEGIN
    SELECT * FROM emp
    WHERE empno = eid;
END //
DELIMITER ;
```

Execute if : CALL getEmployeeById (7521);

How does it works?

1. The value 7521 is passed into the eid parameter
2. The WHERE condition checks empno = eid
3. Only the matching employee is returned.

The imp point is that eid is a parameter, while empno is a column in the table.

2. OUT : Output parameter, the procedure calculat a value and returns it

Ex: Return the total no of employees

```
DELIMITER //
CREATE PROCEDURE totalEmployees (OUT total INT)
BEGIN
    SELECT COUNT(*) INTO total
    FROM emp;
END //
DELIMITER ;
```


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Execute it : CALL totalEmployees (@total);
SELECT @total;

- COUNT(*) : counts the rows in the emp table
- INTO total : stores that count in the procedure -'s output parameter
- @total : is a MySQL session variable that receives the output
- SELECT @total ; : displays the returned value

* total and @total are different. total is the procedure parameter, whereas @total is a session variable.

3. INOUT : Input and output parameter, The procedure receives a value, changes it, and returns the updated value.

Ex : Recive 3000, add 2000, and return 5000

```
DELIMITER //
CREATE PROCEDURE salary (INOUT sal DECIMAL
    (10,2))
BEGIN
    SET sal = sal + 2000;
END//
DELIMITER ;
```


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Execute if : SET @sal = 3000;
CALL salary(@sal);
SELECT @sal;

How to drop and view stored procedures

a. DROP a procedure

DROP PROCEDURE getEmployee;

Removes the stored procedure named getEmploye from the database. This does not delete the employee table

b. SHOW procedure status

SHOW PROCEDURE STATUS;

Displays information about stored procedures, such as their names and databases they belong to

c. SHOW the procedure definition

SHOW CREATE PROCEDURE getEmployees;

Displays the SQL definition used to create getEmployees

NOTE: The procedure must exist before you can drop it or inspect its definition.

Page No.: _____

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